Bremsevæske | HELLA

FunktionBremsesystemer er meget komplicerede, da utallige elementer skal arbejde sammen uden problemer, så bremserne fungerer også i ekstreme situationer. En meget vigtig del er bremsevæsken, for væsken overfører pedalens tryk til bremsesystemet ved hjælp af hydraulisk tryk.For at væsken skal virke korrekt, må dens funktion på intet tidspunkt og under ingen omstændigheder forringes. Bremsevæsken skal derfor være modstandsdygtig over for lave temperaturer, så den ikke bliver tyktflydende eller fryser. Og væsken må heller ikke kunne koge ved høje temperaturer, da dampen danner bobler, hvilket kan forringe bremsevirkningen. Andre vigtige egenskaber for bremsevæsken er korrosionsbeskyttelse, smøring samt forenelighed med forskellige materialer i selve bremsesystemet. Bremsevæsken skal altid være i orden og i tilstrækkelig mængde, så funktionen og trafiksikkerheden garanteres under kørslen. Kogepunkt og viskositetBremsevæsker (DOT 3, DOT 4, DOT 4LV, DOT 5,1) er baseret på en polyglykol-forbindelse, og den er hygroskopisk. Det vil sige, at den optager fugt fra omgivelserne, hvilket som regel sker i form af en diffusion via bremseslangerne eller bremsevæskebeholderen. Jo ældre komponenterne er, desto højere er gennemtrængeligheden. Hvis bremsevæskens vandindhold bliver for højt, falder kogepunktet. Derfor er det tørre kogepunkt, det våde kogepunkt samt viskositeten for en bremsevæske utroligt vigtige.Tørt kogepunkt:Det tørre kogepunkt er kogepunktet for en ny bremsevæske fra en forseglet beholder uden vandindhold.Vådt kogepunkt:Det våde kogepunkt er det kogepunkt, som opnås for en bremsevæske, der indeholder 3,5 % (vægtprocent) vand under bestemte betingelser. Viskositet:Viskositet er et mål for, hvor tyktflydende en væske er. Jo højere viskositeten er, desto mere tyktflydende (mere sej) er en væske; jo lavere viskositeten er, desto mere tyndtflydende (letflydende) er væsken.De fleste væskers viskositet aftager med stigende temperatur. For bremsevæsker, der er afstemt til moderne biler med ESP og ABS (DOT 4 LV og DOT 5.1), er viskositeten meget vigtig, for her er der brug for en relativ høj flydeevne selv ved lave temperaturer.Klassificering iht. DOTFor at bremsevæsker skal overholde de sikkerhedsrelevante krav for disse kriterier, er der udfærdiget bestemte minimumsstandarder. For at de kan

overholdes, findes der en internationalt forpligtende klassificering iht. DOT ("United States Department of Transportation"), som fabrikanterne skal overholde.DOT-klassificeringDOT 3DOT 4DOT 4 LVDOT 5,1Tørt kogepunkt>205 (229)>230 (268)>230 (267)>260 (269)Viskositet ved 100°C [mm²/s] (1,95) (2,34) (2,1) (2,16)Viskositet ved -40°C [mm²/s] (1200) (1315) (675) (810)(I parentes: Testresultater for HELLA bremsevæske)ServiceanvisningHELLA anbefaler følgende skifteintervaller for at sikre bedst mulig bremsefunktion og maksimal sikkerhed:DOT 3: for hver 12 månederDOT 4, DOT 4 LV, DOT 5.1: for hver 24 månederBrug kun bremsevæsker, som er angivet i servicehæftet eller på bremsevæskebeholderens dæksel i køretøjet.Bremsesystemet samt især bremseslangerne bør kontrolleres regelmæssigt for utætheder i forbindelse med serviceeftersyn.Sortiment og emballageHELLA bremsevæske fås i følgende emballager:TypeArtikelnummerEmballage / literDOT 38DF 355 360-0711,00DOT 48DF 355 360-0010,25DOT 48DF 355 360-0110,50DOT 48DF 355 360-0211,00DOT 48DF 355 360-0315,00DOT 48DF 355 360-04120,00DOT 4 LV8DF 355 360-0511,00DOT 4 LV8DF 355 360-0615,00DOT 5,18DF 355 360-0810,50DOT 5,18DF 355 360-0911,00

Brake systems are complex entities in which many elements must work together perfectly to ensure safe functioning, even in extreme situations. The brake fluid plays a very important role in this, as it transfers the power from the pedals to the brake system by means of hydraulic pressure.

In order for it to fulfill its function, the effectiveness of the brake fluid must never be impeded under any circumstances. This means that the brake fluid needs to be resistant to low temperatures in order to ensure that it does not become viscous, or even freeze. It must never boil at high temperatures, as the resulting vapor bubbles could impair the performance of the brakes. Other important properties of brake fluid include corrosion protection, lubrication and good compatibility with the various materials used in the brake system. Brake fluid must function perfectly and be present in sufficient quantities at all times in order to guarantee that the vehicle functions reliably and remains safe on the road.

Boiling point and viscosity

Brake fluid (DOT 3, DOT 4, DOT 4LV,DOT 5,1) is based on a polyglycol bond, and is hygroscopic. This means that it absorbs humidity from its surroundings, usually by means of diffusion via the brake hose or the expansion tank. The older the components, the higher the permeability. If the water content in the brake fluid is too high, the boiling point will drop. As such, the dry boiling point, wet boiling point and viscosity of a brake fluid are crucial.

Dry boiling point:

The dry boiling point is the boiling point of new brake fluid from a sealed container, with no water content.

Wet boiling point:

The term "wet boiling point" refers to the boiling point of a brake fluid with a water content of 3.5% (by weight) under certain conditions.

Viscosity:

The viscosity is a measure of how easily a fluid flows. The higher the viscosity, the thicker (more slow-flowing) a fluid is; the lower the viscosity, the thinner (more fast-flowing) it is.

The viscosity of most substances reduces as their temperature is increased. Viscosity is particularly important in brake fluids designed for modern vehicles with ESP and ABS (DOT 4 LV and DOT 5.1), as they must be able to flow relatively quickly even at low temperatures.

Classification in accordance with DOT

In order to ensure that brake fluids fulfill the safety requirements for these criteria, certain minimum standards have been defined. In order to ensure that these are observed, the United States Department of Transportation has created the DOT classification, which must be observed by manufacturers.

DOT classification	DOT 3	DOT 4	DOT 4 LV	DOT 5.1
Dry boiling point	>205(229)	>230(268)	>230(267)	>260(269)
Wet boiling point	>140(149)	>155(163)	>155(172)	>180(187)
Viscosity at100°C [mm²/s]	< 1.5(1.95)	< 1.5(2.34)	< 1.5(2,1)	< 1.5(2.16)
Viscosity at-40°C [mm²/s]	<1500(1200)	<1800(1315)	<1800(675)	<900(810)

(In brackets: test results for HELLA brake fluid)

Maintenance information

HELLA recommends the following maintenance intervals in order to guarantee optimum brake performance and maximum safety:

- DOT 3: every 12 months
- DOT 4, DOT 4 LV, DOT 5.1: every 24 months

Only use the brake fluids specified in the service manual or on the cover of the expansion tank of the vehicle in question.

In addition to this, the brake system – and the brake hoses in particular – should be checked for leaks regularly during service inspections.

Product range and containers

HELLA brake fluid is available in the following containers:

Type	Article number	Container/liters
DOT 3	8DF 355 360-071	1.00
DOT 4	8DF 355 360-001	0.25
DOT 4	8DF 355 360-011	0.50
DOT 4	8DF 355 360-021	1.00
DOT 4	8DF 355 360-031	5.00
DOT 4	8DF 355 360-041	20.00
DOT 4 LV	8DF 355 360-051	1.00
DOT 4 LV	8DF 355 360-061	5.00
DOT 5.1	8DF 355 360-081	0.50
DOT 5.1	8DF 355 360-091	1.00

Important safety note

Technical information and practical tips have been compiled by HELLA in order to provide professional support to vehicle workshops in their day-to-day work. The information provided on this website is intended for use by suitably qualified personnel only.

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